

ripples, if any, present in the regulated supply. LED1 acts as the power indicator and R2 limits the current through LED1. Regulated 5V DC supply is used for the microcontroller section. Unregulated supply (12V) is used for the LED strips and ULN2803 ICs.

Construction

An actual-size, single-side PCB for microcontroller-based dynamic display using LED strip is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Use IC bases for microcontroller AT89C51 and driver array ULN2803. Before inserting the IC, check proper supply voltage.

Construct 'EFY' — a three-character word for display — using SMD LED strips as shown in Fig. 3. Normally, a 1-metre long LED strip consists of about 60 SMD LEDs with 20 segments or sections. There are three LEDs per section and each section has two points marked as '+' and '-' for connecting the power supply or control signal. Cut out each section as per the requirement to make the word 'EFY.'

For example, for letter 'E,' cut some sections from the 1-metre LED strip, join them and connect between pin 11

PARTS LIST

Semiconductors:	
IC1	- AT89C51 microcontroller
IC2-IC5	- ULN2803 peripheral driver array
IC6	- 7805, 5V regulator
BR1	- 6A bridge rectifier
LED1	- 5mm LED
Resistors (all 1/4-watt, ± 5 per cent carbon):	
R1	- 10-kilo-ohm
R2	- 470-ohm
Capacitors:	
C1	- 10 μ F, 16V electrolytic
C2, C3	- 33pF ceramic disk
C4	- 1000 μ F, 25V electrolytic
C5	- 0.1 μ F ceramic disk
Miscellaneous:	
X1	- 230V AC primary to 12V, 5A secondary transformer
S1	- Push-to-on tactile switch
X _{LED}	- 12MHz crystal LED strip

of IC3 and 12V unregulated supply. Similarly, cut letters 'F' and 'Y' from the strip and connect them to pins 12 and 13 of IC3, respectively.

When the circuit is switched on, 'EFY' text glows in various patterns.

This circuit can also be used for lighting decorations or display during festivals or special occasions. Cut the LED strip into various segments and form the text as per your requirement. Connect each segment to the output of ULN2803 and 12V unregulated supply rail as shown in the circuit diagram.

Note that the PCB layout given here has the unregulated power supply connected to CON3 only, which is connected to port P1 of IC1. If you

want to use ports P0, P2 and P3 of IC1, separate unregulated power supply shown within the dotted line in Fig. 2 has to be provided externally to each of CON2, CON4 and CON5, respectively.

An output pin of ULN2803 can drive one complete LED strip. So eight LED strips can be connected to eight outputs of a ULN2803 IC to form a big character for the display.

Software

The source program (strip.c) for microcontroller-based dynamic display using LED strip is written in 'C' language. It is compiled using the Keil μ Vision4 compiler to generate the hex code. The generated hex code is burnt into the microcontroller using a suitable programmer.

The source program allows for lighting effects like flashing in zig-zag, increasing, decreasing and random patterns. Different delay combinations make the lighting effect more attractive. Function rand() is used to generate the random lighting effects.

EFY note. The source code of this article is available on www.efymag.com website.

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STRIP.C

```
#include <reg51.h>
#include <stdlib.h>
int rand (void);
void Delay(int time);
void main(void) // Main Function
{
    int k, i, num, ran;
    P0 = 0xFF;
    P1 = 0xFF;
    P2 = 0xFF;
    P3 = 0xFF;

    while(1) // Repeat forever infinitely
    {
        for( k= 0; k<5; k++) // on-off
        {
            P0 = 0xFF;
            P1 = 0xFF;
            P2 = 0xFF;
            P3 = 0xFF;
            Delay(50);
            P0 = 0x00;

            P1 = 0x00;
            P2 = 0x00;
            P3 = 0x00;
            Delay(50);

            P0 = num;
            P1 = num;
            P2 = num;
            P3 = num;
            Delay(25); // Delay
            num = num >> 1;
        }
        num = 0x01;
        for ( k = 0; k < 8; k++)
            //backward

            P0 = num;
            P1 = num;
            P2 = num;
            P3 = num;
            Delay(50);
            num = num << 1;
        }
        for( k= 0; k<5; k++) //zig-zag
        {
            P0 = 0x55;
            P1 = 0x55;
            P2 = 0x55;
            P3 = 0x55;
            Delay(50);
            P0 = 0xAA;
            P1 = 0xAA;
            P2 = 0xAA;
            P3 = 0xAA;
            Delay(50);
        }
    }
}
```